

Supplementary Information

Repeatable station terms redistribute long-period response spectra across Japanese strong-motion sites

Haoyu Zhou and Qiang Ma

Supplementary Methods

Record selection and direct periods

The public J-SHIS/NIED `sub1-v2024` flatfile contains 333,808 K-NET and KiK-net records with moment magnitude at least 5 and shortest fault distance no greater than 300 km[1, 2]. Its site table codes ground-surface installations as 1 and borehole installations as 2. We retain code 1 for the residual model and response-spectrum propagation. MF2013 uses the maximum norm of two horizontal oscillator responses, equivalent to RotD100[3]. RotD100 spectral acceleration is read directly at 0.1, 0.2, 0.3, 0.5, 1.0, 2.0, 3.0 and 5.0 s. The residual sample also requires positive spectral acceleration, distance and AVS30; finite magnitude and model inputs; and a crustal, interplate or intraplate MF2013 source class. This screen retains 222,664 surface records at every period. No response coordinate is copied or interpolated.

RotD50 residuals are decomposed on the identical sample as a response-component sensitivity. The RotD50 and RotD100 fields are compared station by station, and record-level RotD50/RotD100 ratios quantify the change in absolute coordinate. Zhao 2006 uses RotD50 because OpenQuake defines the model for the geometric mean of two horizontal components and the flatfile does not provide that exact response coordinate.

Residual model boundary

The primary prediction contains the MF2013 magnitude-distance backbone and the published D1400 and AVS30 terms[4, 5]. D1400 is the depth to the 1.4 km s^{-1} shear-wave velocity horizon. AVS30 is the average shear-wave velocity in the upper 30 m. The anomalous-intensity sensitivity uses the public northeastern and southwestern geographic and depth rules[6]. Incomplete plate-membership metadata restrict this term to a sensitivity calculation. The Philippine Sea Plate term is constant across stations for a qualifying event at a given period and is absorbed by the event fixed effect. The fitted global intercept is retained as a model offset; formal assessment of MF2013 calibration lies outside this design.

Fixed effect decomposition

For residual r_{ij} at station i during event j , the additive model is

$$r_{ij} = \mu + \delta E_j + \delta S_i + \epsilon_{ij}. \quad (1)$$

Alternating grouped least squares estimates δE_j and δS_i . Record-count-weighted event and station means are constrained to zero, so μ contains the global offset. Iteration stops when the largest parameter change falls below $10^{-10} \log_{10}$ units. All 24 period-model combinations converge; the largest final change is 9.99×10^{-11} and the largest iteration count is 1,161.

Event group repeatability

The 1,737 usable earthquakes are assigned to five groups with a fixed random seed. The same partition is used at every period and for both ground-motion backbones. Each fold fits the two-way event-station decomposition separately to four training groups and the held-out group. Paired train and test station vectors are centred with their respective record counts. A station requires at least 15 training records and five test records. The comparison uses the train-test station correlation and the test-record-weighted root mean squared error relative to a zero station term. Each fold contains 347 or 348 test events, and every training and test decomposition converges below $10^{-10} \log_{10}$ units.

Spatial prediction

Five spatial blocks are produced by k -means clustering of standardised longitude and latitude. The site-only model uses shallow-velocity averages, sedimentary-depth variables, elevation, sensor depth and network. The site-and-location model adds station coordinates and two volcanic-front distances. Both histogram gradient-boosting models use fixed hyperparameters: learning rate 0.04, 400 iterations, 15 maximum leaf nodes, 35 minimum samples per leaf and L2 regularisation 0.1. Numeric variables are median-imputed within training folds; network is mode-imputed and one-hot encoded. Predictions are generated only for the held-out block and are centred to a record-weighted zero mean before spectrum propagation.

Alternative ground-motion-model calculation

We evaluate the Zhao et al. 2006 active-crustal, interface and intraslab equations with OpenQuake Engine 3.25.1[7, 8]. The flatfile source class selects the equation branch. Moment magnitude, hypocentral depth and crustal rake are read from the source table; documented shortest fault distance is supplied as rupture distance, and AVS30 is supplied as VS30. RotD50 is used as the closest orientation-independent coordinate to the model’s geometric-mean target. The same surface observations, periods, station threshold, event folds and spatial folds are used for the MF2013 and Zhao calculations. This changes the ground-motion backbone while retaining the data and validation design.

Transfer and event-influence tests

Cross-network tests train on K-NET and evaluate KiK-net, then reverse the direction. A common-feature model removes VS20, VS30 and network identity because these fields have different completeness in the two networks. It retains VS10, AVS30, sedimentary depths, elevation, sensor depth, coordinates and volcanic-front distances. Fixed macroregions are defined by reproducible latitude and longitude boundaries. Event influence is measured after deleting each of the ten events with the largest 3.0 s record count and after deleting all ten at every period. A record-weighted mean shift aligns each perturbed station field with the full field before comparison.

Path and source stratification

Records are grouped into four hypocentral-bearing sectors, four shortest-fault-distance bins and three source classes. Bearing is the initial great-circle direction from the source horizontal coordinates to the station. Stratification can disconnect the bipartite station-event graph, especially within 50 km. Each graph component is solved separately by sparse least squares. Stations require ten records within a stratum, retained components require ten supported stations, and every component is aligned to the full station field by its record-weighted mean difference. Across all 88 period-stratum fits, the maximum relative normal-equation residual is 1.90×10^{-9} and every solver returns an accepted convergence code.

Five random, disjoint event-half comparisons at 1.0, 2.0 and 3.0 s measure sampling reliability within each stratum. A station requires five records in each half. Both half-fields are component-aligned to the full-field reference before comparison. Equal-stratum sensitivity targets are formed by averaging the aligned station terms across at least two available strata, with equal weight per stratum. The primary spatial-validation procedure is then repeated for azimuth-, distance- and source-class-balanced targets; prediction errors retain station record-count weights.

Empirical prediction intervals

For each held spatial block, signed prediction errors from the other four blocks provide record-weighted 5th and 95th percentiles. These quantiles are added to the held-block point prediction. Coverage is computed without using the held-block station terms for interval calibration. Lower and upper station terms are converted to multipliers and propagated to the matched response spectra. The resulting interval describes

station-model prediction error under this spatial split. Source, ground-motion-model and J-SHIS logic-tree uncertainties are outside its scope.

Reference condition conversion

J-SHIS response-spectrum maps provide ordinates on engineering bedrock with $V_S = 400 \text{ m s}^{-1}$ [9, 10]. The station-surface factor replaces the MF2013 shallow-soil term for 400 m s^{-1} with the term evaluated at the station AVS30:

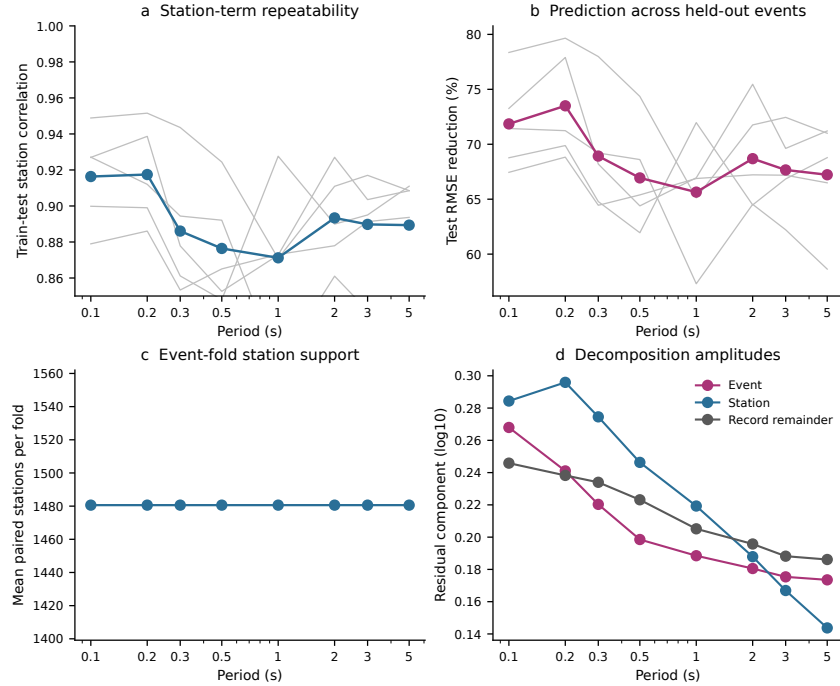
$$F_{\text{surf},i}(T) = 10^{G_s(\text{AVS30}_i, T) - G_s(400, T)}. \quad (2)$$

The cross-validated station factor $10^{\widehat{\delta S}_i(T)}$ is applied after this conversion. All eight period coordinates and four 50-year exceedance probabilities are read from the official map files. The calculation is restricted to matched station cells; national-grid interpolation lies outside this analysis.

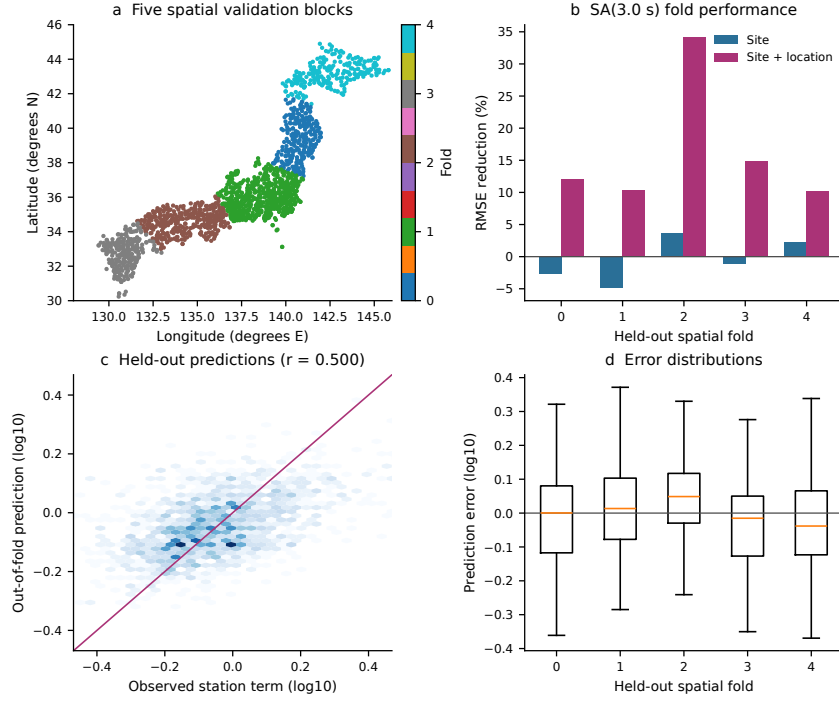
KiK-net waveform check

The waveform check uses 460 paired surface and downhole records from two KiK-net earthquakes[11]. The first set contains 288 converted miniSEED pairs and the second contains 172 original NIED ASCII pairs converted with the scale factors in their headers. Each trace is demeaned and detrended. Horizontal peak amplitude is calculated from the two horizontal components. Surface-to-downhole Fourier amplitude ratios are evaluated from 0.2 to 20 Hz. This check is independent of the station residual model.

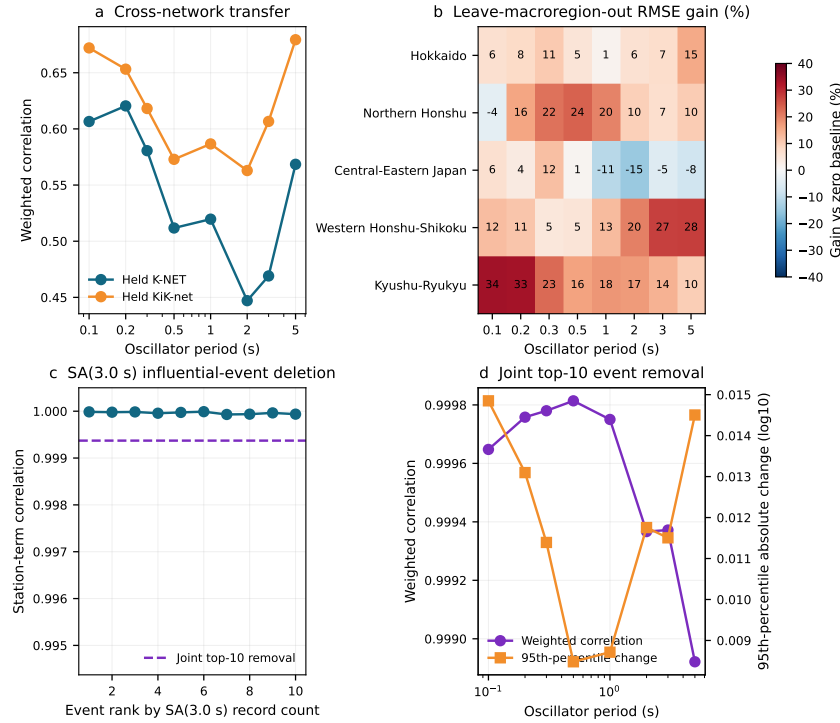
Supplementary Figures



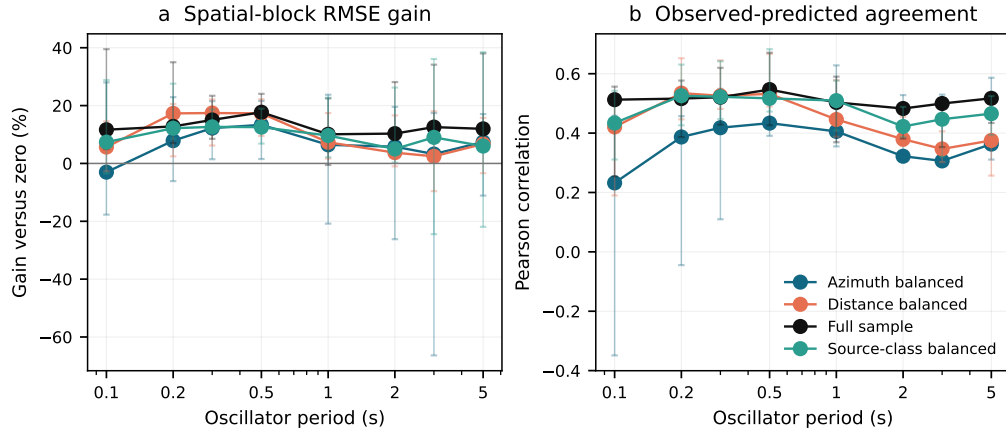
Supplementary Figure 1. Station term repeatability across held-out events. **a**, train-test station correlations for five event folds; grey lines show individual folds and the blue line shows their mean. **b**, test root mean squared error reduction relative to a zero station term. **c**, mean number of stations meeting train and test record thresholds. **d**, event, station and record-remainder amplitudes from the primary MF2013 site residual decomposition.



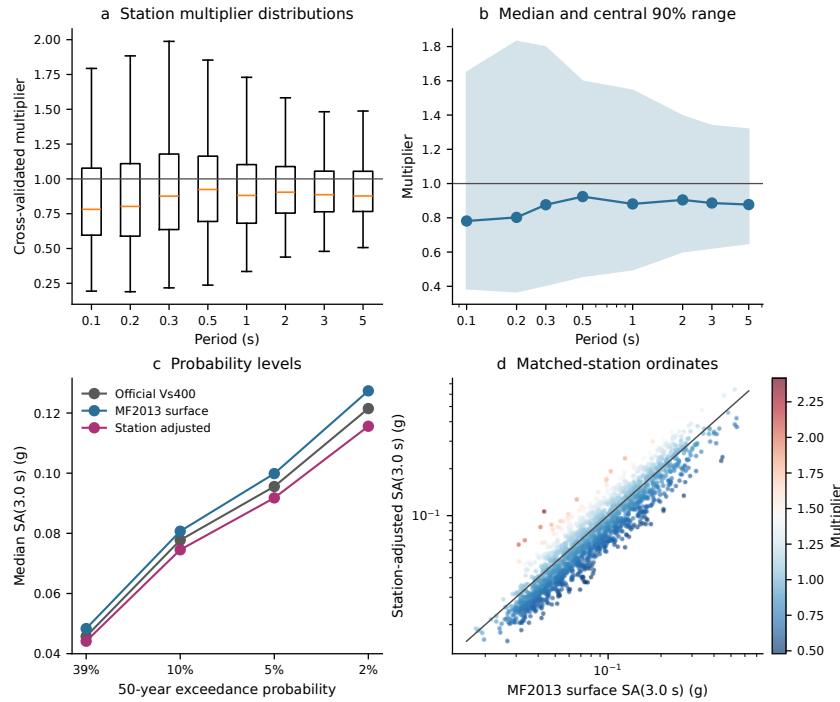
Supplementary Figure 2. Spatial validation of SA(3.0 s) station prediction. **a**, five station blocks used for cross-validation. **b**, root mean squared error reduction in each held-out block for the site-only and site-plus-location models. **c**, observed event-adjusted station terms and out-of-fold site-plus-location predictions. **d**, prediction-error distributions by spatial fold; boxes show interquartile ranges and whiskers extend to 1.5 times the interquartile range.



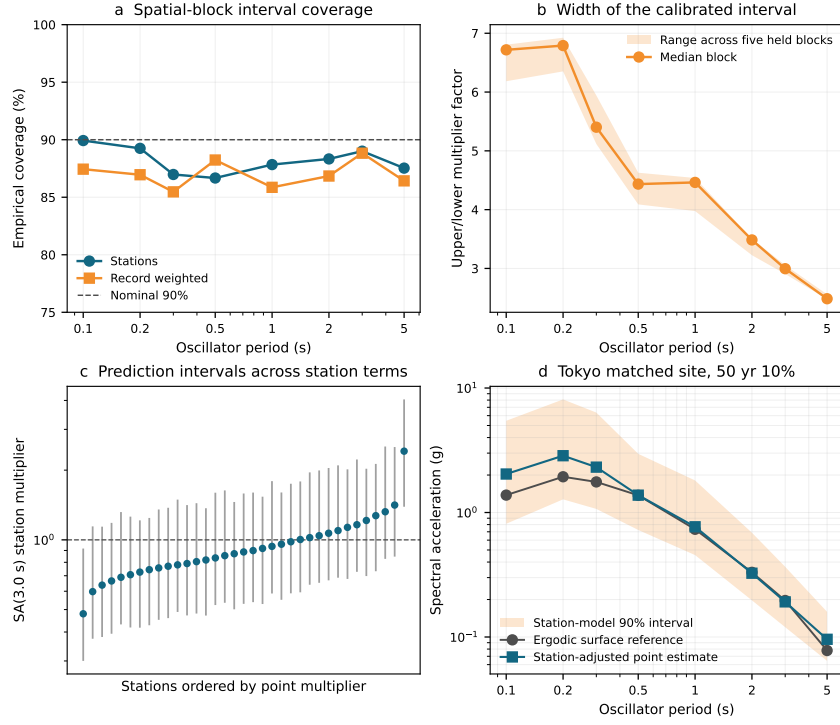
Supplementary Figure 3. Network, geographic and event-influence stress tests. **a**, common-feature transfer between K-NET and KiK-net. **b**, common-feature root mean squared error gain when each fixed macroregion is held out. **c**, SA(3.0 s) station-term correlation after deleting each of the ten highest-count events; the dashed line shows their joint deletion. **d**, station-term correlation and weighted 95th-percentile absolute change after jointly deleting the ten events at every period.



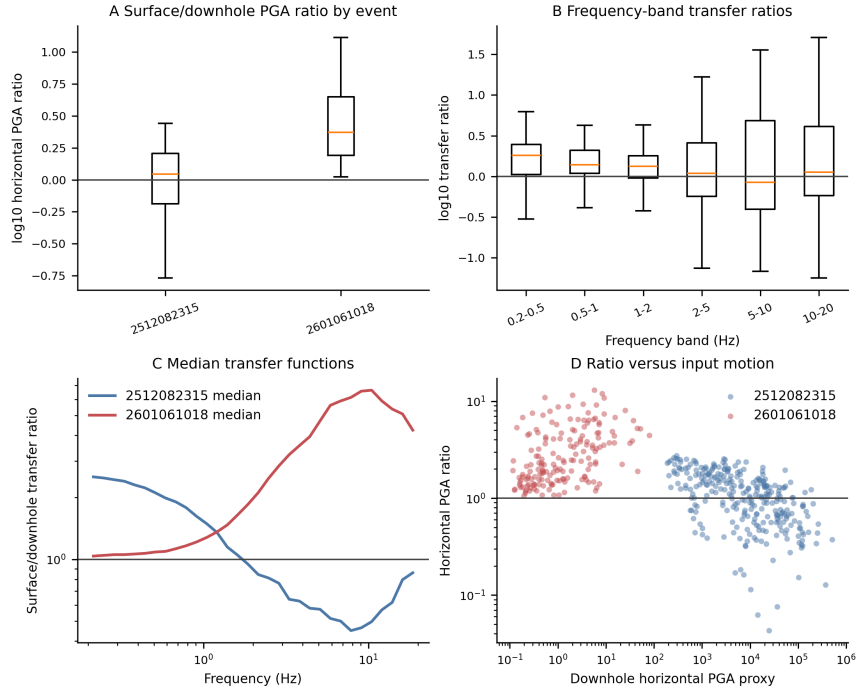
Supplementary Figure 4. Prediction of equal-stratum station fields. **a**, overall spatial-block root mean squared error gain relative to zero. **b**, observed-predicted correlation. Vertical bars span the five held-block results; circles show pooled results. The full-sample target is included for reference.



Supplementary Figure 5. Matched-station response-spectrum sensitivity. **a**, cross-validated station multiplier distributions at the eight direct periods. **b**, median and 5th–95th percentile multiplier range. **c**, median SA(3.0 s) ordinates at four 50-year exceedance probabilities for the official $V_S = 400 \text{ m s}^{-1}$ map, the MF2013 surface projection and the station-adjusted surface projection. **d**, paired MF2013 surface and station-adjusted SA(3.0 s) values at the 50-year 10% level; colour denotes the station multiplier.



Supplementary Figure 6. Spatial-block station-model uncertainty. **a**, empirical coverage of the nominal 90% interval by period. **b**, median interval factor span and its range across the five held blocks. **c**, point multipliers and empirical intervals for 35 ordered SA(3.0 s) stations. **d**, surface-reference and station-adjusted spectra with the station-model interval at the Tokyo-nearest matched station. The interval excludes source, backbone and J-SHIS logic-tree uncertainty.



Supplementary Figure 7. KiK-net surface and downhole waveform comparison. **a**, horizontal surface-to-downhole peak acceleration ratios by earthquake. **b**, frequency-band Fourier amplitude ratios. **c**, median transfer functions for the two event sets. **d**, horizontal peak acceleration ratio against the downhole input-motion proxy. Ratios are shown in log₁₀ units in panels **a** and **b**.

Supplementary Tables

Supplementary Table 1. Primary residual decomposition

Period (s)	Records	Global offset	Event SD	Station SD	Remainder RMSE
			log ₁₀ units		
0.1	222,664	0.284	0.268	0.284	0.246
0.2	222,664	0.215	0.241	0.296	0.238
0.3	222,664	0.172	0.220	0.275	0.234
0.5	222,664	0.106	0.199	0.246	0.223
1.0	222,664	0.017	0.188	0.219	0.205
2.0	222,664	-0.039	0.181	0.188	0.196
3.0	222,664	-0.085	0.175	0.167	0.188
5.0	222,664	-0.134	0.174	0.144	0.186

Supplementary Table 2. Event group station repeatability

Period (s)	Train-test correlation	Test RMSE reduction (%)	Mean paired stations	Mean test events
0.1	0.916	71.9	1,481	347.4
0.2	0.917	73.5	1,481	347.4
0.3	0.886	68.9	1,481	347.4
0.5	0.876	66.9	1,481	347.4
1.0	0.871	65.6	1,481	347.4
2.0	0.893	68.7	1,481	347.4
3.0	0.890	67.7	1,481	347.4
5.0	0.889	67.2	1,481	347.4

Supplementary Table 3. Zhao 2006 ground-motion-model sensitivity

Period (s)	MF2013–Zhao correlation	Median absolute difference (log ₁₀)	Zhao event correlation	Zhao spatial RMSE gain (%)
0.1	0.953	0.064	0.940	16.9
0.2	0.921	0.093	0.941	20.8
0.3	0.919	0.081	0.913	18.9
0.5	0.923	0.073	0.909	20.8
1.0	0.889	0.094	0.915	22.9
2.0	0.801	0.144	0.948	32.1
3.0	0.770	0.151	0.950	37.7
5.0	0.697	0.125	0.946	39.2

Supplementary Table 4. Spatially held-out station prediction

Period (s)	Site-only model RMSE gain (%)	Site and location RMSE gain (%)	Observed-predicted correlation
0.1	-5.2	11.7	0.513
0.2	4.3	12.8	0.516
0.3	10.3	15.0	0.521
0.5	9.0	17.7	0.547
1.0	4.6	10.1	0.503
2.0	0.0	10.3	0.483
3.0	-2.3	12.6	0.500
5.0	-3.5	12.0	0.517

Supplementary Table 5. SA(3.0 s) network and fixed-macroregion transfer

Held group	Full-model RMSE gain (%)	Common-feature RMSE gain (%)	Common-feature correlation
K-NET	4.1	8.7	0.469
KiK-net	21.2	20.5	0.607
Hokkaido	7.2	6.9	0.284
Northern Honshu	5.4	6.8	0.304
Central-Eastern Japan	6.2	-5.2	0.402
Western Honshu–Shikoku	28.9	27.5	0.386
Kyushu–Ryukyu	13.4	13.8	0.372

Supplementary Table 6. Joint deletion of the ten highest-count events

Period (s)	Station-term correlation	Weighted 95th-percentile absolute change ($\log_{10}$)
0.1	0.99965	0.0148
0.2	0.99976	0.0131
0.3	0.99978	0.0114
0.5	0.99981	0.0085
1.0	0.99975	0.0087
2.0	0.99937	0.0118
3.0	0.99937	0.0115
5.0	0.99892	0.0145

Supplementary Table 7. SA(3.0 s) path and source stratification

Stratification	Stratum	Paired stations	Full-field correlation	Split-half correlation	95th percentile difference
Hypocentral azimuth	000–090°	938	0.280	0.877	0.474
Hypocentral azimuth	090–180°	448	0.678	0.762	0.236
Hypocentral azimuth	180–270°	1,089	0.777	0.967	0.317
Hypocentral azimuth	270–360°	1,345	0.886	0.945	0.168
Fault distance	0–50 km	138	0.644	0.563	0.289
Fault distance	50–100 km	680	0.496	0.604	0.427
Fault distance	100–200 km	1,515	0.974	0.951	0.080
Fault distance	200–300 km	1,213	0.963	0.968	0.092
Source type	Crustal	1,494	0.942	0.905	0.120
Source type	Interplate	960	0.934	0.947	0.073
Source type	Intraplate	1,379	0.924	0.902	0.131

Supplementary Table 8. Equal-stratum spatial prediction sensitivity

Target field	Period (s)	Stations	Overall RMSE reduction (%)	Overall correlation	Minimum fold reduction (%)
Azimuth balanced	1.0	1,220	6.4	0.406	-20.8
Azimuth balanced	2.0	1,220	5.8	0.322	-26.2
Azimuth balanced	3.0	1,220	3.2	0.306	-66.4
Distance balanced	1.0	1,285	7.3	0.446	1.7
Distance balanced	2.0	1,285	3.7	0.379	-1.0
Distance balanced	3.0	1,285	2.5	0.347	-9.6
Source balanced	1.0	1,369	9.6	0.509	2.2
Source balanced	2.0	1,369	5.0	0.422	0.3
Source balanced	3.0	1,369	9.0	0.447	-24.5

Supplementary Table 9. Matched-station 50-year 10 percent response spectra

Period (s)	Station multiplier			Official V_S400 median spectral acceleration	MF2013 surface spectral acceleration (g)	Station adjusted acceleration (g)
	5th	Median	95th			
0.1	0.386	0.781	1.650	0.946	0.978	0.817
0.2	0.368	0.802	1.830	1.068	1.154	0.936
0.3	0.407	0.876	1.797	0.914	1.022	0.891
0.5	0.458	0.924	1.597	0.652	0.737	0.668
1.0	0.498	0.881	1.544	0.315	0.341	0.308
2.0	0.602	0.904	1.395	0.132	0.140	0.130
3.0	0.624	0.886	1.338	0.078	0.081	0.075
5.0	0.651	0.877	1.318	0.030	0.031	0.028

Supplementary Table 10. Empirical station-model prediction intervals

Period (s)	Station coverage (%)	Record-weighted coverage (%)	Median upper/lower multiplier factor
0.1	89.9	87.4	6.72
0.2	89.3	87.0	6.79
0.3	87.0	85.5	5.40
0.5	86.7	88.2	4.43
1.0	87.8	85.9	4.32
2.0	88.3	86.8	3.49
3.0	89.0	88.8	3.00
5.0	87.5	86.4	2.48

Supplementary Table 11. City-nearest SA(3.0 s) station cases

City	Station	Distance (km)	Multiplier	Change (%)	Surface SA	Adjusted SA (g)
Tokyo	TKY007	2.4	0.975	-2.5	0.196	0.192
Osaka	OSK005	3.6	0.885	-11.5	0.223	0.197
Sapporo	HKD180	8.6	1.312	+31.2	0.124	0.163
Sendai	MYG013	5.0	0.890	-11.0	0.131	0.116
Fukuoka	FKOH03	14.1	0.748	-25.2	0.035	0.026

Supplementary Table 12. Response-component sensitivity

Period (s)	Station-field correlation	Weighted difference RMSE	95th percentile absolute difference	Median record RotD50/RotD100
0.1	0.9995	0.0076	0.0179	0.853
0.2	0.9995	0.0081	0.0183	0.839
0.3	0.9994	0.0079	0.0176	0.834
0.5	0.9993	0.0073	0.0177	0.831
1.0	0.9990	0.0077	0.0197	0.826
2.0	0.9988	0.0074	0.0194	0.820
3.0	0.9986	0.0071	0.0180	0.819
5.0	0.9983	0.0064	0.0177	0.820

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